

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Revision Test 2026

MATHEMATICS — SSC-I (9th Grade) — ANSWER KEY (Version 16)

Syllabus: Unit 1 - Real Numbers, Unit 2 - Logarithms, Unit 4 - Factorization and Algebraic Manipulation, Unit 5 - Linear Equations and Inequalities, Unit 7 - Coordinate Geometry

Total Marks: 43

Time Allowed: 2 Hours

Version 16

SECTION A — Answers to Multiple Choice Questions ($24 \times 1 = 24$ Marks)

Q1. In $3^5 = 243$, the number **243** is called the:

Correct answer: **B) value** 1 mark

In an expression of the form $a^n = b$, **a** is the base, **n** is the exponent (or index) and **b** is the **value**. So in $3^5 = 243$, the base is 3, the exponent is 5 and the value is 243.

Source in her textbook: Unit 1, §1.5 Exponents / Indices — SLO: recall base, exponent and value

Q2. $(a^5 \times a^{-3})^2 \div a^{-2}$ simplifies to:

Correct answer: **C) a^6** 1 mark

$a^5 \times a^{-3} = a^{5-3} = a^2$. Then $(a^2)^2 = a^4$. Finally $a^4 \div a^{-2} = a^{4-(-2)} = a^6$.

Source in her textbook: Unit 1, §1.5 Laws of Exponents — SLO: apply the laws of exponents to simplify expressions with real exponents

Q3. On the number line, $\sqrt{19}$ lies between the two consecutive integers:

Correct answer: **A) 4 and 5** 1 mark

$16 < 19 < 25$, so $\sqrt{16} < \sqrt{19} < \sqrt{25}$, that is $4 < \sqrt{19} < 5$. On the number line $\sqrt{19}$ is therefore located between **4 and 5** (its value is about 4.36).

Source in her textbook: Unit 1, §1.2 Real Numbers on the Number Line — SLO: depict real numbers on the number line

Q4. The multiplicative identity in the set of real numbers is:

Correct answer: **D) 1** 1 mark

For every real number a , $a \times 1 = 1 \times a = a$. So **1** is the multiplicative identity. (0 is the *additive* identity, since $a + 0 = a$.)

Source in her textbook: Unit 1, §1.3 Properties of Real Numbers — SLO: know the properties of real numbers

Q5. The Babylonian system of numerals was a sexagesimal system, that is, a system with base:

Correct answer: **C) 60** 1 mark

The textbook states that the Babylonian system of numerals, which was based on the earlier Sumerian system, was a **sexagesimal** system, i.e. a system with **base 60**. The concept is still in use today in measuring time and angles.

Source in her textbook: Unit 1, opening section — Babylonians' Contribution to the Numeral System — SLO: recall the history of numbers

Q6. If $\log x = 2.6180$, then the value of x (found from the antilog table) is:

Correct answer: **B) 415.0** 1 mark

The characteristic is 2, so the number has 3 digits before the decimal point. From the antilog table, antilog of the mantissa .6180 is 4.150, and shifting the decimal point according to characteristic 2 gives $x = 415.0$.

Source in her textbook: Unit 2, §2.5 Antilogarithm — SLO: give the concept of antilog and use tables to find the antilog of a number

Q7. $\log_7 7$ is equal to:

Correct answer: **A) 1** 1 mark

By the definition of a logarithm, $\log_a b = c$ means $a^c = b$. Here $7^1 = 7$, so $\log_7 7 = 1$. In general $\log_a a = 1$ and $\log_a 1 = 0$.

Source in her textbook: Unit 2, §2.2 Logarithm of a Number to the Base a — SLO: define logarithm of a number to the base a

Q8. $\log 15 - \log 3$ is equal to:

Correct answer: **D) $\log 5$** 1 mark

By the quotient law of logarithms, $\log p - \log q = \log (p/q)$. So $\log 15 - \log 3 = \log (15/3) = \log 5$. Note that it is *not* $\log 12$; subtracting logs corresponds to dividing the numbers, not subtracting them.

Source in her textbook: Unit 2, §2.6 Laws of Logarithm (quotient law) — SLO: prove and apply the four basic laws of logarithm

Q9. Written in standard notation, 3.06×10^{-4} is:

Correct answer: **C) 0.000306** 1 mark

A negative exponent means the decimal point moves to the **left** by that many places. Moving the point in 3.06 four places to the left gives **0.000306**.

Source in her textbook: Unit 2, §2.1 Scientific Notation and Standard Notation — SLO: express a number in standard form of scientific notation and vice versa

Q10. The characteristic of $\log 3456$ is:

Correct answer: **B) 3** 1 mark

For a number greater than 1, the characteristic is **one less than the number of digits** before the decimal point. 3456 has 4 digits, so the characteristic is $4 - 1 = 3$. (The mantissa is the decimal part read from the log table.)

Source in her textbook: Unit 2, §2.3 Characteristic and Mantissa — SLO: define a common logarithm, characteristic and mantissa of the log of a number

Q11. The factors of $27x^3 + 64$ are:

Correct answer: **A) $(3x + 4)(9x^2 - 12x + 16)$** 1 mark

This is a **sum of cubes**: $27x^3 + 64 = (3x)^3 + (4)^3$. Using $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ with $a = 3x$ and $b = 4$ gives $(3x + 4)(9x^2 - 12x + 16)$. **Watch the signs:** the bracket sign is + and the middle sign inside the trinomial is -. (For $a^3 - b^3$ it is $(a - b)(a^2 + ab + b^2)$.)

Source in her textbook: Unit 4, §4.3 Factorization of $a^3 \pm b^3$ — SLO: factorize expressions of the type $a^3 + b^3$

Q12. $3x + 3y + ax + ay$ factorizes to:

Correct answer: **D) $(x + y)(3 + a)$** 1 mark

Group the terms: $(3x + 3y) + (ax + ay) = 3(x + y) + a(x + y) = (x + y)(3 + a)$. This is the type $ac + ad + bc + bd = (c + d)(a + b)$.

Source in her textbook: Unit 4, §4.1 Factorization of $ac + ad + bc + bd$ — SLO: recall factorization of expressions of the type $ac + ad + bc + bd$

Q13. $x^3 + 6x^2 + 12x + 8$ factorizes to:

Correct answer: **B) $(x + 2)^3$** 1 mark

Compare with $a^3 + 3a^2b + 3ab^2 + b^3 = (a + b)^3$, taking $a = x$ and $b = 2$: $x^3 + 3(x^2)(2) + 3(x)(4) + 8 = x^3 + 6x^2 + 12x + 8$. Hence the factors are **$(x + 2)^3$** .

Source in her textbook: Unit 4, §4.3 Factorization of $a^3 + 3a^2b + 3ab^2 + b^3$ — SLO: factorize expressions of that type

Q14. $x^2 + 10x + 25 - y^2$ factorizes to:

Correct answer: **A) $(x + 5 + y)(x + 5 - y)$** 1 mark

The first three terms form a perfect square: $x^2 + 10x + 25 = (x + 5)^2$. So the expression is $(x + 5)^2 - y^2$, a difference of two squares, which factorizes as **$(x + 5 + y)(x + 5 - y)$** .

Source in her textbook: Unit 4, §4.1 Factorization of $a^2 + 2ab + b^2 - c^2$ — SLO: recall factorization of that type

Q15. The complete factorization of $81m^4 - 16n^4$ is:

Correct answer: **C) $(9m^2 + 4n^2)(3m + 2n)(3m - 2n)$** 1 mark

$81m^4 - 16n^4 = (9m^2)^2 - (4n^2)^2 = (9m^2 + 4n^2)(9m^2 - 4n^2)$. The second bracket is itself a difference of two squares, so it factorizes further into $(3m + 2n)(3m - 2n)$. The **complete** factorization is therefore $(9m^2 + 4n^2)(3m + 2n)(3m - 2n)$. Option A is correct but not complete.

Source in her textbook: Unit 4, §4.1 Factorization of $a^2 - b^2$ — SLO: recall factorization of expressions of the type $a^2 - b^2$

Q16. The factors of $3x^2 + 10x + 8$ are:

Correct answer: **D) $(3x + 4)(x + 2)$** 1 mark

For $ax^2 + bx + c$ with $a = 3$, $b = 10$, $c = 8$, find two numbers whose product is $ac = 24$ and whose sum is 10: these are 6 and 4. Then $3x^2 + 6x + 4x + 8 = 3x(x + 2) + 4(x + 2) = (3x + 4)(x + 2)$.

Source in her textbook: Unit 4, §4.2 Factorization of $ax^2 + bx + c$ — SLO: factorize expressions of the type $ax^2 + bx + c$

Q17. Which of the following is **NOT** a linear equation in one variable?

Correct answer: **C) $x^2 = 4$** 1 mark

A linear equation in one variable can be written as $ax + b = 0$, $a \neq 0$, and the exponent of the variable is always 1. The textbook states that **$x^2 = 4$ is non-linear because the exponent of the variable is not 1.** (An equation such as $xy = 14$ is also non-linear, because it involves a product of variables.)

Source in her textbook: Unit 5, §5.1 Linear Equation in One Variable (Key Points 1 and 3) — SLO: recall linear equation in one variable

Q18. If $a < b$, then $a + c < b + c$ for every real number c . This is the:

Correct answer: **A) additive property of inequality** 1 mark

Adding the same number to both sides of an inequality does not change the direction of the inequality. This is the **additive property**. Trichotomy states that exactly one of $a < b$, $a = b$, $a > b$ holds; the transitive property states that if $a < b$ and $b < c$ then $a < c$.

Source in her textbook: Unit 5, §5.4 Properties of Inequalities — SLO: recognize properties of inequalities (trichotomy, transitive, additive and multiplicative)

Q19. The solution of the inequality $(2x/3) - 1 < 3$ is:

Correct answer: **B) $x < 6$** 1 mark

$(2x/3) - 1 < 3 \Rightarrow 2x/3 < 4$ (additive property) $\Rightarrow 2x < 12$ (multiply both sides by 3, a positive number, so the sign does not reverse) $\Rightarrow x < 6$.

Source in her textbook: Unit 5, §5.5 Solving Linear Inequalities with Rational Coefficients — SLO: solve linear inequalities with rational coefficients

Q20. If $\sqrt{3x - 2} = 4$, then x is equal to:

Correct answer: **D) 6** 1 mark

Square both sides to reduce the radical equation to linear form: $3x - 2 = 16 \Rightarrow 3x = 18 \Rightarrow x = 6$. Check: $\sqrt{3(6) - 2} = \sqrt{16} = 4$. $\checkmark$ A common slip is to forget to square the 4 (giving $3x - 2 = 4$, $x = 2$) or to square only one side (giving $3x = 16$, $x = 16/3$).

Source in her textbook: Unit 5, §5.2 Equations Involving Radicals — SLO: reduce equations involving radicals to simple linear form and find their solutions

Q21. The abscissa of every point lying on the y-axis is:

Correct answer: **C) zero** 1 mark

The abscissa is the **x-coordinate** of a point. Every point on the y-axis has the form $(0, y)$, so its abscissa is **zero**. (Similarly, the ordinate of every point on the x-axis is zero.)

Source in her textbook: Unit 7, §7.1 Conventions for Coordinates in the Cartesian Plane; Miscellaneous Exercise 7 Q1 — SLO: use conventions for coordinates in the Cartesian plane

Q22. If the ordered pairs $((a + 1)/2, b - 3)$ and $(4, 5)$ are equal, then the value of a is:

Correct answer: **B) 7** 1 mark

Two ordered pairs are equal only when their corresponding elements are equal. So $(a + 1)/2 = 4 \Rightarrow a + 1 = 8 \Rightarrow a = 7$, and $b - 3 = 5 \Rightarrow b = 8$.

Source in her textbook: Unit 7, Miscellaneous Exercise 7 Q1 (equality of ordered pairs) — SLO: use conventions for coordinates in the Cartesian plane

Q23. If B is the mid-point of the line segment AC , then $AC : BC$ is:

Correct answer: **D) 2 : 1** 1 mark

If B is the mid-point of AC then $AB = BC$ and $AC = 2 BC$. Therefore $AC : BC = 2 : 1$.

Source in her textbook: Unit 7, §7.3 Mid-point Formula; Miscellaneous Exercise 7 Q1 — SLO: derive the mid-point formula and calculate the mid-point of a line segment

Q24. The distance between the points $(-6, 3)$ and $(6, 8)$ is:

Correct answer: **A) 13 units** **1 mark**

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{[(6 - (-6))^2 + (8 - 3)^2]} = \sqrt{(12^2 + 5^2)} = \sqrt{(144 + 25)} = \sqrt{169} = \mathbf{13 \text{ units.}}$$

Source in her textbook: Unit 7, §7.2 Distance Formula — SLO: find distance between two points in the plane

Q	1	2	3	4	5	6	7	8	9	10	11	12
Key	B	C	A	D	C	B	A	D	C	B	A	D
Q	13	14	15	16	17	18	19	20	21	22	23	24
Key	B	A	C	D	C	A	B	D	C	B	D	A

SECTION B — Answers to Short Questions ($3 \times 3 = 9$ Marks)

Q.2(i)(a) Simplify, using the laws of exponents: $(x^5 \times x^{-2}) \div x^4$ **1 mark**

$$x^5 \times x^{-2} = x^{5+(-2)} = x^3 \quad (\text{product law: } a^m \cdot a^n = a^{m+n})$$

$$x^3 \div x^4 = x^{3-4} = x^{-1} = \mathbf{1/x} \quad (\text{quotient law}) \quad \mathbf{1 \text{ mark}}$$

Q.2(i)(b) Evaluate: $(81)^{3/4}$ **1 mark**

$$81 = 3^4, \text{ so } (81)^{3/4} = (3^4)^{3/4} = 3^{4 \times 3/4} = 3^3 = \mathbf{27} \quad \mathbf{1 \text{ mark}}$$

Q.2(i)(c) Write $\sqrt[4]{a^7}$ in exponential form. **1 mark**

$$\sqrt[n]{a^m} = a^{m/n}. \text{ The index of the radical becomes the denominator of the exponent, so } \sqrt[4]{a^7} = \mathbf{a^{7/4}}.$$

1 mark

Source in her textbook: Unit 1, §1.4 Radicals and Radicands, §1.5 Laws of Exponents — SLOs: transform an expression given in radical form to exponential form and vice versa; apply the laws of exponents.

Q.2(ii) Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$, evaluate $\log 48$, showing every step and naming the law used. **3 marks**

Step 1. Write 48 as a product of powers of 2 and 3: $48 = 16 \times 3 = 2^4 \times 3$ **1 mark**

Step 2. $\log 48 = \log (2^4 \times 3) = \log 2^4 + \log 3$ (**product law:** $\log mn = \log m + \log n$)
 $= 4 \log 2 + \log 3$ (**power law:** $\log m^n = n \log m$) **1 mark**

Step 3. Substitute the given values:

$$= 4(0.3010) + 0.4771 = 1.2040 + 0.4771 = \mathbf{1.6811} \quad \mathbf{1 \text{ mark}}$$

Marking note: the question asks her to **name the law** at each step. Deduct the second mark if the product and power laws are used correctly but not named. Accept $48 = 2^4 \times 3$ written as $48 = 2 \times 2 \times 2 \times 2 \times 3$.

Source in her textbook: Unit 2, §2.6 Laws of Logarithm — SLOs: prove the four basic laws of logarithm; apply laws of logarithm to convert lengthy processes of multiplication and exponentiation into easier processes of addition.

Q.2(iii) Solve $|2z - 3| = 9$ and verify both solutions. **3 marks**

By the definition of absolute value, $|x| = a$ ($a > 0$) means $x = a$ **or** $x = -a$. So:

$$\text{Case 1: } 2z - 3 = 9 \Rightarrow 2z = 12 \Rightarrow \mathbf{z = 6} \quad \mathbf{1 \text{ mark}}$$

$$\text{Case 2: } 2z - 3 = -9 \Rightarrow 2z = -6 \Rightarrow \mathbf{z = -3} \quad \mathbf{1 \text{ mark}}$$

Verification **1 mark**

$$\text{For } z = 6: |2(6) - 3| = |12 - 3| = |9| = 9 \quad \checkmark$$

$$\text{For } z = -3: |2(-3) - 3| = |-6 - 3| = |-9| = 9 \quad \checkmark$$

Solution set = $\{-3, 6\}$.

Marking note: the third mark is for the **verification**, which the question explicitly asks for. If both roots are correct but no verification is shown, award 2 of the 3 marks.

Source in her textbook: Unit 5, §5.3 Absolute Value; Exercise 5.3 — SLOs: define absolute value; solve the equation involving absolute value in one variable.

SECTION C — Answers to Long Questions ($2 \times 5 = 10$ Marks)

Q.3(i)(a) Factorize completely: $1 + 4x^4$ 2 marks

This is of the type $a^4 + 4b^4$, with $a = 1$ and $b = x$. Add and subtract the middle term needed to complete the square:

$$1 + 4x^4 = 1 + 4x^4 + 4x^2 - 4x^2 \quad (\text{add and subtract } 4x^2) \quad \text{1 mark}$$

$$= (4x^4 + 4x^2 + 1) - 4x^2$$

$$= (2x^2 + 1)^2 - (2x)^2 \quad (\text{a perfect square minus a square})$$

$$= (2x^2 + 2x + 1)(2x^2 - 2x + 1) \quad \text{1 mark}$$

Check: $(2x^2 + 1)^2 - (2x)^2 = 4x^4 + 4x^2 + 1 - 4x^2 = 4x^4 + 1 \checkmark$

Q.3(i)(b) Find the square root of $49p^2 - 84pq + 36q^2$ by factorization. 1 mark

$$49p^2 - 84pq + 36q^2 = (7p)^2 - 2(7p)(6q) + (6q)^2 = (7p - 6q)^2$$

$$\sqrt{49p^2 - 84pq + 36q^2} = \pm(7p - 6q) \quad \text{1 mark} \quad (\text{accept } 7p - 6q)$$

Q.3(i)(c) Simplify: $(x^2 - 64)/(x^2 + 11x + 24) \div (x^2 - 16x + 64)/(x^2 + 3x)$ 2 marks

Step 1 — factorize every expression:

$$x^2 - 64 = (x - 8)(x + 8) \quad | \quad x^2 + 11x + 24 = (x + 3)(x + 8)$$

$$x^2 - 16x + 64 = (x - 8)^2 \quad | \quad x^2 + 3x = x(x + 3) \quad \text{1 mark}$$

Step 2 — change $\div$ into $\times$ by inverting the second fraction:

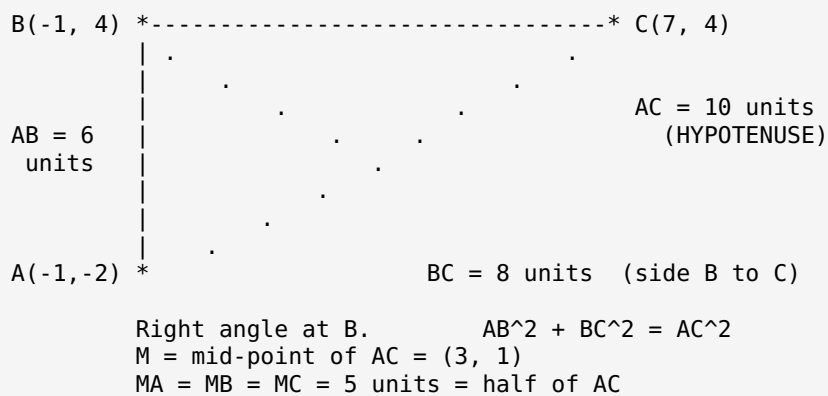
$$= [(x - 8)(x + 8) / ((x + 3)(x + 8))] \times [x(x + 3) / (x - 8)^2]$$

Step 3 — cancel the common factors $(x + 8)$, $(x + 3)$ and one $(x - 8)$:

$$= x / (x - 8) \quad \text{1 mark}$$

Source in her textbook: Unit 4 — §4.3 (type $a^4 + 4b^4$), §4.6 Square Root by Factorization, §4.5 Use of HCF and LCM to reduce fractional expressions involving $+$, $-$, $\times$, $\div$.

Q.3(ii) A(-1, -2), B(-1, 4) and C(7, 4) are the vertices of a triangle. 5 marks



(a) Lengths of the sides, by the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ 2 marks

$$AB = \sqrt{(-1 - (-1))^2 + (4 - (-2))^2} = \sqrt{0 + 36} = \mathbf{6 \text{ units}}$$

$$BC = \sqrt{(7 - (-1))^2 + (4 - 4)^2} = \sqrt{64 + 0} = \mathbf{8 \text{ units}} \quad \mathbf{1 \text{ mark}}$$

$$AC = \sqrt{(7 - (-1))^2 + (4 - (-2))^2} = \sqrt{64 + 36} = \sqrt{100} = \mathbf{10 \text{ units}} \quad \mathbf{1 \text{ mark}}$$

(b) Proof that $\triangle ABC$ is a right triangle 1 mark

In a right triangle the greatest side is the hypotenuse, so here the hypotenuse is **AC**. By the converse of Pythagoras' theorem:

$$AB^2 + BC^2 = 6^2 + 8^2 = 36 + 64 = 100 \text{ and } AC^2 = 10^2 = 100$$

Since $AB^2 + BC^2 = AC^2$, $\triangle ABC$ is a **right triangle**, right-angled at **B**, and **AC is the hypotenuse.** **1 mark**

(c) Mid-point of the hypotenuse and equal distances 2 marks

By the mid-point formula, $M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$ applied to A(-1, -2) and C(7, 4):

$$M = ((-1 + 7)/2, (-2 + 4)/2) = (6/2, 2/2) = \mathbf{M(3, 1)} \quad \mathbf{1 \text{ mark}}$$

Now find the distance from M to each vertex:

$$MA = \sqrt{(3 - (-1))^2 + (1 - (-2))^2} = \sqrt{16 + 9} = \sqrt{25} = \mathbf{5 \text{ units}}$$

$$MB = \sqrt{(3 - (-1))^2 + (1 - 4)^2} = \sqrt{16 + 9} = \sqrt{25} = \mathbf{5 \text{ units}}$$

$$MC = \sqrt{(3 - 7)^2 + (1 - 4)^2} = \sqrt{16 + 9} = \sqrt{25} = \mathbf{5 \text{ units}}$$

Since **MA = MB = MC = 5 units**, the mid-point of the hypotenuse is at an equal distance from all three vertices. Note also that $5 = \frac{1}{2} \times 10$, i.e. this distance is half the length of the hypotenuse. **1 mark**

Source in her textbook: Unit 7 — §7.2 Distance Formula (Example 4, vertices of a right triangle), §7.3 Mid-point Formula, Miscellaneous Exercise 7 (mid-point of the hypotenuse of a right triangle) — SLOs: find distance between two points in the plane; derive the mid-point formula and calculate the mid-point of a line segment; apply distance and mid-point formulas to solve real life situations.

— END OF ANSWER KEY —